Ethics and images in in social media research

by Nicole Taylor, Louie Dean Valencia-García, Angela VandenBroek, Ashley Stinnett, and Alejandro Allen

This study examines strategies for anonymizing online data in a way that follows ethical guidelines while also retaining the fluid nature of participant engagement with social media. We explore unanticipated ethical issues that emerged as we began capturing and anonymizing social media posts for our ethnography of college students and online sociality. Using examples from our data, we illustrate our penciling technique to anonymize images for the naked eye while also considering the digital footprint. We consider methodological and ethical challenges of anonymization in a specific internet research context — user generated social media images obtained with informed consent — and explore the implications of our findings on the potential life of digital content.

Contents

Introduction
Background
Method
Discussion and conclusions

Introduction

Researchers are increasingly incorporating the study of online worlds into research projects addressing identity and sociality. The easy access that researchers have to online data coupled with rapidly changing media technologies highlights the continued importance of attending to methodological and ethical issues in online worlds (Boellstoerff, *et al.*, 2012; Horst and Miller, 2012). For example, when research takes place in a social media context available to an infinite public rather than in local communities where behavior is on display for a much smaller audience, what are the implications for privacy? What does it mean to protect the confidentiality of participants who may understand and value privacy differently than the researcher?

This study examines strategies for anonymizing online data in a way that follows ethical guidelines while also retaining the fluid nature of participant engagement with social media. We explore unanticipated ethical issues that emerged as we began capturing and anonymizing social media posts for our ethnography of college students and online sociality. We present a technique that we are calling penciling, inspired by a method developed by Tiidenberg (2018). Our intention is to explore the boundaries of vulnerability for online images in order to anonymize photos for the naked eye while also considering the digital footprint.

Background

Privacy

New and emerging technologies, such as machine learning and facial recognition, in tandem with rapidly changing ideas about privacy, have complicated the ethical implications of collecting, managing, anonymizing, and sharing online image data. Social media has normalized public exposure, effectively changing values and perceptions associated with privacy and, in turn, communication practices (Gershon, 2010; Horst and Miller, 2012).

Social media is an invitation into an archive of the carefully cultivated self, where concepts of public and private are blurred (Manderson, *et al.*, 2015). Confessional culture in the U.S., rooted at least partially in popular media obsessed with highlighting the private lives of celebrities and everyday people through gossip forums and reality television, has turned the concept of privacy on its head (Hall, 2016; Khamis, *et al.*, 2017).

The question of what constitutes publicly accessible online data remains unclear (Townsend and Wallace, 2016; see also franzke, *et al.*, 2020; Lomborg and Bechmann, 2014; Tiidenburg, 2018; Tromble and Stockman, 2017). Ever changing social media policies complicate the issue of public versus private domains.

For example, users may not be fully aware of privacy policies that they agreed to when creating social media accounts, and they might change their privacy settings partway through a study. This means that a researcher who forgoes the consent process may capture publicly posted social media images that the user later decides to make private.

Moreover, no social media account can truly be private. Even an account with the maximum privacy settings in place still contains identifiers, such as screen names, locations, and profile pictures, which are public by default. Social media posts can be captured and shared by people with access to private data or during periods of public access.

For example, posts on Twitter are commonly screenshotted and reposted as images as a way to “subtweet” and preserve posts that might be later removed. Likewise, archival projects, like the Internet Archive’s *Wayback*

Machine, capture snapshots of the Internet, including some social media posts, as a method of preservation and documentation.

Researchers must also attend to how different platforms shape participants' ideas about privacy (Beninger, *et al.*, 2014; Patterson, 2018). Social media users' conceptions of audience matter because reproducing social media content, even if it is publicly available and in anonymized form, changes the assemblage of viewers (Ravn, *et al.*, 2020).

Research shows that teens and young adults understand and negotiate privacy online differently than older generations of social media users (Hargittai and Marwick, 2016; Marwick and boyd, 2014), highlighting the influence of socio-cultural norms and tacit rules on social media use. While older generations often focus on institutional privacy from government and corporations, younger generations are more likely to focus on social privacy (Raynes-Goldie, 2010).

Social privacy concerns managing access to data based on social relations to prevent context collapse where different social relationships have access to the same data (Davis and Jurgenson, 2014). An example of context collapse would be if you posted an image with your best friend from last night's party, and one of your Facebook friends shared that image with your employer. By extracting image data from the technical and social infrastructures teens and young adults use to manage social privacy, researchers risk introducing new locations of context collapse.

Marwick and boyd (2014) describe a model of networked privacy that is constantly in flux and heavily dependent on a combination of audience, social norms, and technology. They argue that a focus on participants' expectations, sensitivity of information, and vulnerability of participants may be the best measure of risk and harm (Markham and Buchanan, 2015).

Further, contextual integrity, a concept introduced by Nissenbaum (2011), prioritizes attending to norms of informational flow both off-line and online in attending to privacy. Such norms are context dependent rather than universal, requiring that researchers continually revisit the ethical implications of their methods (see also franzke, *et al.*, 2020). Here we see the value of using participant observation coupled with in depth interviews as a methodological avenue to gain greater understanding about these shifting social contexts.

Informed consent, confidentiality, & visual data

Researchers have long debated the ethical implications of gathering individuals' publicly accessible Internet data without permission (see boyd, 2016; boyd and Crawford, 2012; franzke, *et al.*, 2020; Tiidenberg, 2018; Townsend and Wallace, 2016). Specifically, boyd (2016) interrogates the value of academic informed consent procedures, pointing out that participants often sign, without really paying attention to, written or verbal explanations.

Interpretations of images may shift over time with additional data collection so that participants who consented and perhaps even collaborated in a visual production process may not be fully aware of how images will be used (Emmel and Clark, 2011; Heider, 2006). One strategy for mitigating this risk is engaging in an ongoing, collaborative consent process (Pink, 2012).

Privacy is relational, which complicates anonymization, especially when a single social media post can include multiple individuals who may not have consented to participating in the research (Zimmer and Kimmer-Kurlanda, 2017). Social media sites "tend to reflect an aggregate of an individual's private spheres, the previously dyadic contact with each friend or relative co-present in the same space" [1].

Some scholars question, given current technological advances, such as automated facial recognition and GIS locators, whether an assurance of participant anonymity is even possible when it comes to visual research methods (Cox, *et al.*, 2014).

With regard to social media data, the traditional ethnographic practices of using pseudonyms and removing identifying information is not enough to protect confidentiality. Instead, traceability needs to be a key consideration (Beaulieu and Estalella, 2012). Though researchers can reduce the traceability of online data, it is not entirely possible to anonymize this information.

The distance principle examines the distance between online data collected and the individuals who created the data in determining potential for harm (Lomborg, 2013; Markham and Buchanan, 2015, 2012). The closer the distance, the greater the risk for identifying participants. For example, distance between a few identifiable Facebook posts is closer than for 200 tweeted memes.

Images are especially prominent in social media to the extent that "Individuals' experiences are visualized across social media platforms" [2]. In fact, some popular platforms, including Instagram, TikTok, and Snapchat, are visual by design — photos and videos are the focal point of users' profiles.

As such, in social media research, capturing images to complement textual data is a crucial part of understanding how people construct narratives, communities, and identities online (Pearce, *et al.*, 2020). Visual data presents additional, unique ethical and methodological challenges, particularly in terms of anonymization for use in publications.

Researchers use visual methods to capture aspects of daily life, identity, and sociality that they cannot obtain through other methods. As such, altering data to protect confidentiality can undermine its purpose altogether, further complicating the ethical dilemma. Strategies used to anonymize images may render them useless as a form of visual data (Clark, 2012). Standard methods for protecting confidentiality in narrative/textual data generally include changing the names of participants and locations.

When it comes to images, ethnographers have employed a variety of techniques, including blurring faces (see Goodwin, 2006; Mendoza-Denton, 2008) or using pseudonyms alongside identifiable photos of individuals (see Bourgois and Schonberg, 2009; Carney, 2015; De Leon, 2015). Other strategies include replacing social media photos with cartoon-like illustrations (Miller and Sinanan, 2017) or artist renditions (Abidin, 2018).

Next, we explore the challenges of working with online images through a discussion of our recent study that relied heavily on visual data from social media. The process of anonymizing digital images that we describe in this paper emerged out of a much larger ethnographic study exploring online identity, presentation of self, emotional expression, and social norms of online interaction among U.S. college students (Taylor and Nichter, 2022). Our methods for this study included in-depth interviews, focus groups, and long-term participant observation in participants' social media sites.

As part of the participant-observation portion of this ethnographic study, we captured hundreds of posts across Instagram, Snapchat, Twitter, and Facebook, most of which featured identifiable images of participants. The highly stylized social media images that we captured contained rich and nuanced visual data associated with personal style, stance, angles, facial expression, and other embodied identity markers. We needed to find a way to retain aspects of the original images that were crucial to analysis of identity creation online.

Method

Publication, anonymization, and privacy

Upon capturing social media posts of participants, we faced the ethical and technical challenges of anonymizing these images for publications. While narrative accounts can do much to describe our observations, concrete examples of visual elements of the field are necessary to convey the subtlety and complexity of their roles.

In accordance with the university's Institutional Research Ethics Board (IRB) expectations that the researcher makes "reasonable efforts to maintain confidentiality," we had initially considered typical anonymizing strategies, such as blurring faces, blacking out location tags, participant names, and names of friends. These are common techniques employed by anthropologists and others.

We invited a digital historian, Louie Dean Valencia-García, to help us test the veracity of our method for protecting confidentiality. We experimented with the degree of pixilation used to blur faces, from barely softening features to erasing features altogether. Throughout this process, we considered how the blurring might render a face unrecognizable to someone who had never met a given participant, but would still leave the person recognizable to someone who knows the participant.

For example, when we blurred faces, Valencia-García, who neither participated in data collection nor had access to unedited social media images, could not identify participants. However, those of us who participated in data collection were able to recognize participants because we had met them and seen their social media posts. Yet, when we tried a high level of pixilation, it erased facial features altogether, resulting in an expressionless skin mask, which rendered the images useless as exemplars of our analysis.

We found that moderate pixilation blurred faces beyond recognition to a stranger, but still revealed features, skin color, and expressions. A further drawback to this approach was that even moderate blurring erased nuanced aspects of facial features, a crucial element for the study of online identity construction and emotional expression.

This anonymization method had an ethical drawback as well. The digital historian — who often thinks about the repercussions and challenges of long-term digital data management — pointed out that it would be incredibly difficult to guarantee anonymization of any digital image. Most academics in the United States today are aware of digital services, such as Turn-It-In, which use machine learning to compare student submissions to large databases of submitted material, public search engines, and the like in order to identify similarities in content and style that could indicate cheating.

Some digital humanities scholars use similar text analysis technology, such as Voyant, to search and analyze large digital troves, made possible by the creation of new digital archives of print material that were previously difficult to locate without visiting a special collection. Using this type of technology, one scholar had located previously unknown works by nineteenth century poet Walt Whitman by searching keywords in the *ProQuest Historical Newspapers* database to recognize Whitman's distinctive style (Blalock and Farrar, 2019).

Collaborations between computer science and art history have led to the production of similar machine learning technologies for visual media such as classifying painting genres (Arora and Elgammal, 2012; Elgammal, *et al.*, 2018) and identifying undiscovered works of artists (Cömert, *et al.*, 2021).

Today, through digital technology, users are constantly creating information. It is impossible to predict future applications of data and ease of identification through textual and visual analysis. In particular, this is a challenge for those scholars who, for ethical reasons, want to anonymize data.

A digital historian, who might want to use such technology to identify someone for the historical record, would consider this type of technology a boon. However, for an ethnographer bound by IRB guidelines for maintaining confidentiality, such technology represents an ethical challenge.

Testing the efficacy of common techniques

This experience led us to more in-depth questions about the efficacy of common anonymization techniques for visual data given these new and emerging technologies. We explored ways in which digital technology could facilitate the ability for the casual Internet user to access widely available facial recognition technology to identify a participant in a digital image.

For the present study, we relied on raster images in RGB (Red, Green, Blue) color formats, such as JPEG. Raster files store image data in a matrix of pixels arranged in columns and rows. Computer screens display images using additive color, which means red, green, and blue light are added together to make more than 16 million colors.

For the machine, these colors appear as values between zero and 255 for the three light values (*e.g.*, a bright red would be 255, 0, 0 and a pure white would be 255, 255, 255) when stored in eight-bit JPEG files. Thus, each digital image is a unique series of numerical values.

However, just as the human eye is capable of perceiving these collections of tiny boxes of light as faces, machines can use these values to identify mathematically patterns of color, contrast, and shape that, given enough examples, can teach the machine to "see" faces too. Like Whitman's turn-of-phrase, every face is unique, and the more digitized/digital images of a person that appear online, the more data that is available for comparison.

If a person has a large social media following, appears in popular publications, or simply posts photos of themselves across multiple social media platforms, the chances increase for their data to be collected and for them to be identifiable despite a researcher's attempts to anonymize that data.

Since 2011, Google Images has provided a reverse image search feature. Users can upload an image to Google's system to search for similar images. Google converts the image into a mathematic algorithm by analyzing an image's features [3].

Other search engines, such as Bing, TinEye, and Yandex, offer similar services. Suggested content features on apps like Pinterest and image repositories like Getty Images, Flickr, and Shutterstock also use these types of machine learning technologies.

When you upload an image into Google Image, the search engine generates a query. The search engine then compares that uploaded image to billions of images in its databases using convolutional neural networks, a form of machine learning inspired by biological neurons and the visual cortex processing of imagery. These images can be photos uploaded as profile pictures or posts to social media platforms, such as Twitter, Facebook, Instagram, or other Web sites. The analysis of an image compares pixels for similar patterns. Other popular reverse image search services, like Yandex, TinEye, and Bing, use similar internally developed machine learning software to analyze images and rank results.

To attempt to preserve anonymity, a researcher might believe that blurring facial features or covering a participant's eyes with black bars using programs such as Photoshop can be effective. Edits such as these essentially make a person unidentifiable to most human eyes, as humans tend to focus on the foreground of images to recognize socially significant patterns and rarely pay attention to small background details.

Machines, however, compare the pixels left intact in the image after blurring or blacking out the socially significant data, such as a tree's leaf pattern in the background or the color and pattern of a person's shirt. Machines have the computational power to attend quickly to all details equally, regardless of social significance.

For example, an image taken using an iPhone 7, released in 2016, takes photos at 3,024 x 4,032 pixels, and thus has 12,192,768 pixels potentially available for comparison. Even taking into account pixels that become unusable by obscuring someone's face in a photo, Google Image's software can still find similarities based upon pixels that remain available for comparison. These pixel patterns functionally work as digital fingerprints.

Additionally, most photos taken with phones or digital cameras by default hold "metadata" called EXIF data, which include information such as geographical location, date and time of the photograph, and the specifics of the

device and its settings. EXIF data makes it easier to identify a person.

Most social media sites, however, now remove most or all EXIF data on upload to protect the privacy of users, including Facebook, Twitter, Instagram, and Snapchat. This is not always the case, however. Notoriously, following the 6 January 2022 riots in Washington, D.C., investigators discovered that Parler, the social media platform preferred by rioters, did not delete EXIF data, allowing authorities and Internet sleuths to identify rioters and prove their presence at the U.S. Capitol.

To test the efficacy of our strategies, we developed a "Google Image Search Test" to determine whether participants in our study would be identifiable. To model the test, we located images of a popular Spanish exercise and fitness trainer, Sergio Peinado, who, at the time, had more than 1,985,000 followers on YouTube.

We used both de-identifying strategies, where we blurred photos and added black bars across Peinado's eyes, effectively rendering the image unusable to describe facial features and emotional expression. We then ran those images through the reverse search feature at http://images.google.com.

Google immediately identified the social media influencer in all de-identified trials, despite the anonymization techniques we had used. This indicated that traditional anonymization strategies for visual print data were, in fact, not effective in an online context.

We wondered whether it would be this easy to identify social media photos of our research participants, none of whom were Internet famous. We ran the Google Image Search Test on images captured from 60 participants' social media accounts and created a spreadsheet, systematically testing to see if each participant could be identified using the Google Image Search test.

When possible, we selected three publicly available images from each participant, with preference for individual photos, or photos with only two to three people. Given that the anonymization (blurring or placing bars) was not effective in the first test, we left these images unedited, with the primary goal being to see whether Google Image could identity participants.

Most participants passed the test (*i.e.*, were not identified), and only six failed. One person failed due to a number of theater headshots they had made available on various Internet sites. This participant went on to perform in a prominent Broadway show in New York after graduating, and so his Internet presence was especially robust. Another participant failed because the Instagram photo that we tested appeared as their profile picture for a public group forum.

The other four failed because the photos we pulled from their Instagram or Facebook accounts appeared as their public profile pictures on LinkedIn (in one case) or Twitter (in three cases). For Twitter, any picture used as a person's profile picture (no matter how popular the person's Twitter profile is) will appear through the Google Image Search Test. LinkedIn offers multiple privacy options for profile photos, including first-degree connections only, one's LinkedIn network, all LinkedIn members, and anyone on or off LinkedIn. This latter option results in profile photos being indexed by Google and other search engines. This does not seem to be the case for Facebook, Instagram, or Snapchat [4].

We ran the same test on several other search engines with similar reverse image services, including Bing, TinEye, and Yandex. Results were similar across Google and Yandex. For example, any photos that were Twitter profile pics or appeared on public Internet sites appeared through search tests on Yandex just as they did on Google. Bing and TinEye only identified the participant whose theater headshot appeared on a number of public Internet sites.

Table 1: Results of reverse image testing.

Reason for photo match	Google	Yandex	TinEye	Bing
Twitter Profile Photo	3	3	0	0
LinkedIn Profile Photo	1	0	0	0
Instagram Profile Photo	0	1	0	0
Instagram Post	0	1	0	0
Photo on public Web site	2	2	1	1

Two additional participants failed the reverse image search test when we expanded it to include these additional search engines. Yandex was able to match photos for these two participants and provide a link to their public Instagram accounts, a phenomenon that was unique to this search engine. For one participant, the photo that we tested was their Instagram profile picture, but for the other participant, the photo matched an Instagram post buried in their feed from several years ago. Both participants were social media influencers at the time of our study with paid sponsorships from companies, and they each have many followers — roughly 5,000 and 1,500 respectively.

To explore this finding in Yandex further, we tested five other participants' profile pictures from their public Instagram accounts. All five participants had fewer than 1,000 followers, and Yandex did not match any of their Instagram profile pictures. For these Instagram accounts with relatively low follower counts, Yandex instead matched the photos with Instagram profile pictures of other people who looked similar and had a more prominent social media profile. In these cases, Yandex matched a participant's Instagram profile picture with a celebrity or social media influencer with similar facial features, clothing, hairstyle and color, and/or who posed similarly in the photo. This led us to wonder if the size of a person's Instagram following influences the accuracy of reverse image searches of their profile pictures on Yandex.

Bing and TinEye were unable to generate possible matches for any photo we uploaded to the site except for one — the participant who failed the Google Image Search Test due to his theater headshot appearing on a number of Web sites. This same headshot appears as his Twitter profile picture, but these two search engines only matched it to a theater Web site. Because of the widespread, public nature of this theater headshot photo, all search engines easily identified this participant.

In two instances, Yandex matched photos with old profile pictures from Twitter, or even picked up an image embedded in someone else's Twitter feed. When we clicked on the link to go to their Twitter pages, one was a dead link; the other displayed a different profile picture. In these cases, it appeared that Yandex had cached old images from Twitter that had been deleted. As a result, these two participants could not be identified based on the match.

Although the search engines were unable to identify most participants, the blurring and black out techniques described earlier still left participants vulnerable, especially if those not identified at the time of our test later posted more public images for comparison — particularly as they move into adulthood and post public professional headshots for employment. This test revealed vulnerabilities in online image privacy that could not be adequately addressed through traditional anonymization techniques.

Given the importance of social privacy to the generation that we studied, this is a particularly problematic issue as it introduces the possibility of context collapse with only a quick online image search. Thus, these methods were

not effective for protecting participants.

Penciling: A tactic for machine vision

In a second attempt at de-identification, we drew on an anonymization technique highlighted by Tiidenberg (2018), which involved editing images to make them appear as pencil sketches. Markham developed a method for creating composite accounts of individuals, events, or interactions from representative elements of the data. Ethical fabrication allows the researcher to present a fictional narrative that cannot be traced back to specific individuals (Markham, 2012).

Tiidenberg (2018) adapted the ethical fabrication method for visual data by using filters in an IOS application to render the photos as pencil sketches to make participants less recognizable while retaining important visual details. Inspired by Tiidenberg's description of this approach, we examined the efficacy of this kind of image alteration and developed a method using Adobe Photoshop [5] that we call penciling. We found that our penciling method, which could be adapted to accommodate features of individual images, reduced technical traceability and retained visual data.

To pencil a image, we used a series of filters and blending modes in Photoshop to alter every pixel in a image, while preserving edges within a image. First, two copies of an image were created as layers. Both images were desaturated to remove all color, and the second image was inverted to turn whites (255, 255, 255) to black (0, 0, 0), blacks to white, and invert grey values (*e.g.*, a light gray of 200, 200, 200 becomes a dark gray of 55, 55, 55).

We then set the inverted copy to the color dodge blend mode. This mode divides the RGB color values of the bottom layer (a) by the inverse of the top layer (b) — that is a/(255-b). As the two copies have exact opposite values, this produces a completely white image where all values equal 255. We then used a blur filter on the inverted layer to alter each pixel's color value.

The color dodge formula resulted in brighter areas of the original image appearing washed out to solid white and areas where there was higher contrast between pixels of the original image appearing as black or shades of grey. Minimal blurring resulted in thin lines only in areas of high contrast, while substantial blurring resulted in shaded areas where there were mid-tones in the original image.

This method allows you to edit images along a visual spectrum that, on one end, clearly retains the visual quality of the photo, and on the other end, transforms the photo into the thin lines and subtle shading that one associates with a pencil sketch. By adjusting the blur intensity, we were able to control the level of detail in the penciling result.

You can reintroduce color to the image by placing a copy of the original color photo in a third layer in the "color" blend mode, which replaces gray pixels with the color values of the original image. This process alone alters the color value of every pixel in the image as well as significantly changes the contrast between pixels — thus fundamentally changing the way machines "see" the image.

Depending on risk and privacy needs, you could also alter facial features and other elements of the photograph by using the paint tool on any part of the inverted layer with white to render that area white. This would be particularly important for any lettering or words in the photograph.

While most machine learning image recognition is based on comparison of patterns of pixel data, some, including Google's reverse image search, can "read" words within images and use them to find similar images that also contain those words. Thus, leaving identifiable words within the image could increase the risk of identifying the source image.

We chose to retain the color quality and facial features from captured images to convey subtle aspects of identity and style in the original posts. Figure 1 displays a sketch with fine, thin lines and Figure 2 illustrates a slightly more realistic photo effect. In both images, we used Adobe Photoshop to alter every pixel, making them equally unidentifiable in terms of their digital footprint.

Figure 1: Photo edited to resemble a pencil sketch with fine, thin lines.







Figure 2: Photo edited to retain visual quality of the original image.

Both versions decrease recognizability to the naked eye, but to varying degrees. Determining the level of the effect depended on the quality of the photo captured from social media as well as the degree of editing applied by the participant prior to posting it to social media. Social media platforms have image specifications, including size and file type; images are compressed and converted accordingly upon upload, which affects quality.

Further, moderately or heavily edited photos or those taken with inexpensive camera phones were more difficult to manipulate on our end, as they tended to appear grainy and washed out in places. In most cases, we had to operate within a very narrow range to retain the important features of the photo.

Even then, we found that some areas of the image may blur out completely, blending with nearby pixels. Without knowing what exactly participants had done to manipulate their social media photos, we worked to salvage image quality on our end by experimenting with other editing tools, such as contrast and opacity. For example, Figure 3 was incredibly difficult to pencil. We did not know exactly which editing techniques the participant had used on the photo, but when we applied the penciling technique, all of the lines were crisp and defined except for her face, which appeared blurry. In particular, her cheekbones blurred into the rest of her face initially. After hours of manipulating the blurring intensity and contrast, we finalized the image, which provides enough definition of the participants face that the audience can discern her expression.

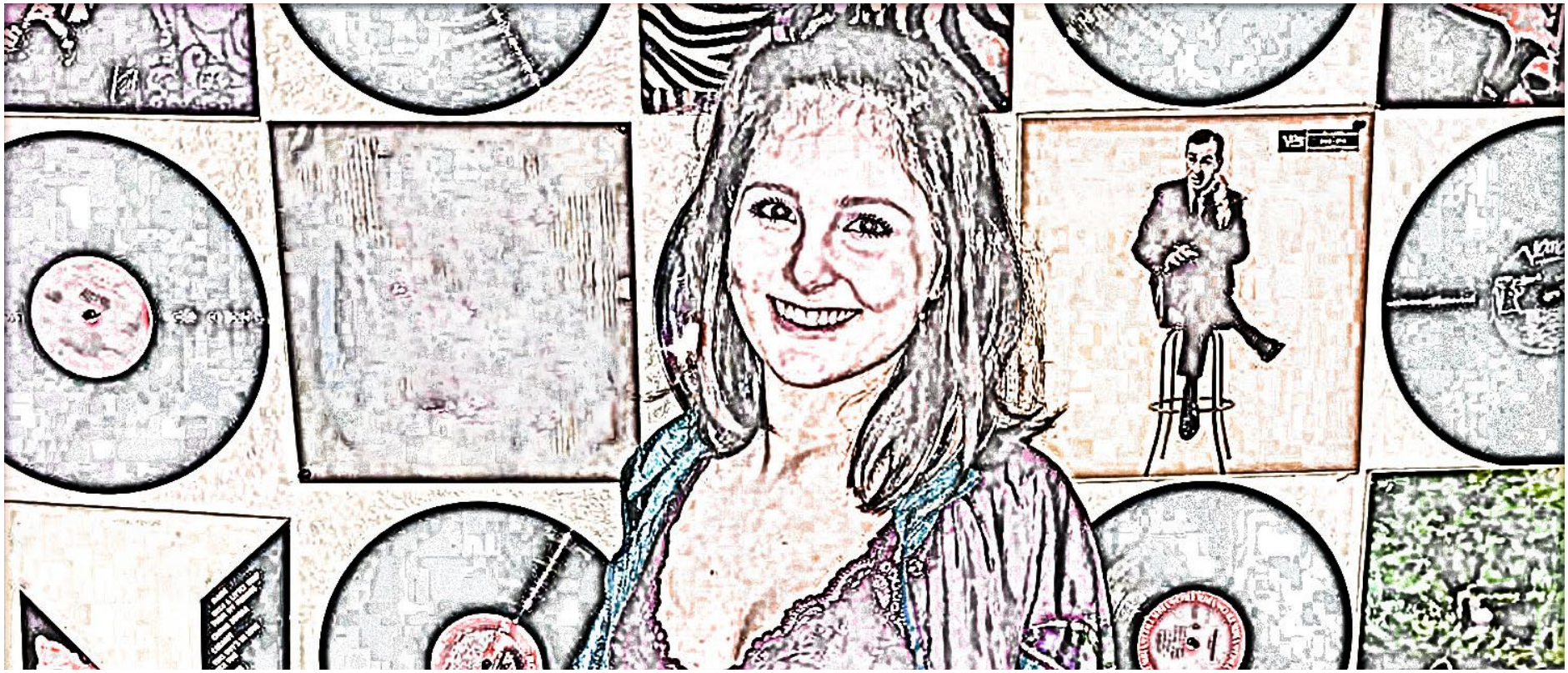




Figure 3: Photo edited with attention to blurring contrast and intensity to retain facial features.

Similarly, Figure 4 was difficult to pencil as well due to editing of the original photo. This participant described how he edited his photo during an in-depth interview, which provided some insight into how penciling is affected by filters and editing techniques. He told us that he used multiple filters on the image, including face softening and sepia tone. Additionally, he applied a filter that blurred the background while keeping the foreground image crisp and clear, and he added the milk carton, pink sky, clouds and stars to the image using Photoshop. As a result, various areas of the photo are grainy, blurry and washed out.







Figure 4: Photo edited to retain details of facial expression at the expense of background imagery.

Figures 3 and 4 both contained text in the background that could have potentially identified participants through a reverse image search. We obscured the text by using the spot healing brush, which uses an algorithm to sample areas near the selected area and replaces the text with pixel data similar to its surroundings.

Retaining visual quality while penciling photos proved challenging, but we were able to retain aspects of the image, such as stance, posture, facial features, and expressions while decreasing recognizability for both the naked eye and digital footprint. Ultimately, we had to weigh the benefits and drawbacks of our method to determine its effectiveness.

Using a photograph of a family member of one of our authors, we penciled the photograph and uploaded it to her personal Google Photos storage account. Unlike Google Images and other search engines with image matching capabilities, Google Photos is able to use a curated collection of personal photos to create highly specific and accurate search results on personal data, including searching by individuals appearing in the photograph collection.

Despite having access to over 9,000 identified photographs of this family member to compare to, our altered penciled image was unidentifiable as the individual or as a potential match for the individual by Google Photos. To test our method systematically, we uploaded all altered pencil images of research participants into the reverse image search features of Google, Bing, TinEye, and Yandex. In every case, images were unidentifiable.

Though our penciled images were not visually as attractive as the original photos, they met the criteria for our research needs by ensuring online digital anonymity. We found our penciling method effective in protecting the identities of our participants while also maintaining the fluid nature of youth engagements with social media.

Discussion and conclusions

As ethnographers, we work hard to build rapport with and gain the trust of people that we study so that they will share their inner worlds with us. However, modern technology, in tandem with rapidly changing conceptions of and values associated with privacy, has complicated the ethical implications of collecting, managing, anonymizing, and sharing online image data. As such, building and maintaining trust requires that ethnographers conducting Internet research engage in continual, clear communication with participants about data collection, management, and dissemination of findings.

In the context of a rapidly changing technological context, ethnographers who work with human subjects must remain vigilant about potential risks associated with visual data, especially as studies focused on social media use are on the rise. Early on, our team of anthropologists worked with the university's IRB to determine anonymization methods to protect the confidentiality of our research subjects.

Initially, based on IRB guidelines and widely accepted anonymization practices used by researchers in our discipline, we felt confident that blurring faces and blacking out names and location tags on social media posts we captured would ensure confidentiality of our participants.

Collaboration between the anthropologists and a digital historian revealed how easy it is to identify anonymized images captured from social media accounts. In short, the anthropologists on the team learned that we did not know what we did not know, and the consequences for our knowledge limitations could be potentially significant for research subjects.

This experience underscores not only the value of interdisciplinary collaboration, but also the importance of seeking out and being open to perspectives and expertise that scholars in other disciplines have to offer.

It is difficult to predict what new technologies will develop. As such, we must assume it is possible to identify a person through a reverse image search. Currently, most academic journals publish digitally. If journals publish digital images of subjects that have not altered every pixel in the original photo, subjects may be identifiable.

For the purposes of our study, the most feasible and effective strategy for protecting confidentiality was penciling. This method, similar to methods used by other digital media researchers (Miller and Sinanan, 2017; Tiidenberg, 2018), provides the added benefits of complex controls over the process and the ability to retain facial features and expressions, key elements for interpreting and illustrating image production on social media.

Recent trends have moved toward giving participants more autonomy and agency of their narrative through methods such as digital storytelling. Scholars give participants cameras to collect data and tell their own stories. It is worth considering the risks associated with such public exposure of research subjects. For example, it is possible that data collected by the participants could be used against them at some point by a potential employer or in a court of law. In the U.S., it is possible to subpoena research data as evidence against a participant (Boelstorff, *et al.*, 2012).

Further, participants may not understand the reach and scope of images and videos that the researcher captures and distributes, let alone potential consequences related to technological advances and changing political climates — it is impossible to predict how identification could harm participants in the future (Wiles, *et al.*, 2012).

One potential solution for scholars researching social media to protect confidentiality is to maintain visual data behind paywalls that require verification to access. In this case, scholars can work with institutional libraries to create repositories where verified researchers can ask for access to sensitive images for scholarly purposes, agreeing to conditions that would place restrictions on the use and downloading of those images.

Scholars could also work with academic journals and request that they not digitally publish sensitive photographic material, and instead provide a link to those institutional library's pages for verification.

When working with vulnerable populations, researchers can draw on additional strategies to anonymize images. While it may not be appropriate to publish such images digitally, using them in a research or poster presentation or including them in print-only versions of publications is safer. For example, showing photos and posts in a conference presentation, where the audience can view them for only a minute or two on a projected screen would pose a minimal risk to participants.

In this case, an audience member would have to use their cell phone to take a photo of the image at a distance, making it much more difficult to run a reverse image search online. The effort required combined with the likely size and composition of an academic conference audience would sufficiently protect participants' confidentiality.

Further, engaging in an ongoing, collaborative consent process with participants is an important strategy for guiding the ethical use of images (Pink, 2012). Early on, before we understood the implications of an online image's digital footprint, we anticipated using traditional anonymizing techniques, including face blurring and blacking out names and other identifiers.

These were the strategies that we named in the official consent form and discussed with participants at the onset. However, through the course of the study, as we learned about the ease of identifying online images through a simple Google search, we changed course to decrease recognizability for both the digital footprint and naked eye. This creative problem-solving approach, a concept Warfield, *et al.* (2019) refer to as "ethics on the go," is employed by Internet researchers who must continually adapt to the changing nature of social media.

After data collection concluded, we checked in with participants whose social media images we wanted to publish to ask their permission. We provided them with contextual information, including publication venue, intended audience, and how we planned to anonymize images.

We explained to participants that we planned to edit images to appear as pencil sketches, a process that would alter every pixel in the photo and decrease recognizability, and that we would use a pseudonym in place of their actual name. Most students readily provided permission, some wanted to see images we planned to use and decide on a case-by-case basis, and a few did not grant permission.

The results of our follow-up consent process highlighted the need for an ongoing collaborative consent process (Strand, *et al.*, 2003) and revealed the interpretive nature of informed consent. As boyd explains:

> Ethics aren't a checklist. Nor are they universal ... Reasonable people — even experts — differ on what they think is ethical. And disciplines have different standards for how to navigate ethics. We need ethics to not just be tacked on, but to be an integral part of how everyone thinks about what they study, build, and do. [6]

The Association of Internet Researchers (AoIR) recommends guidelines rather than a one-size-fits-all approach, underscoring the importance of making judgement calls that account for context (franzke, *et al.*, 2020).

The AoIR 3.0 report on Internet ethical guidelines states, "Perhaps most importantly: such judgment is deeply relational and intersubjective. As such, it requires continuous cross-checking, mutual critique, and/or corroboration and expansion — precisely through ongoing dialogue and discussion with peers and trusted colleagues" [7].

For our team of anthropologists, we found that bringing a digital historian on board enriched our understanding of ethics and online images. We also engaged in a series of discussions with the university's IRB, which included faculty from a wide range of disciplines, in order to troubleshoot issues we encountered and obtain feedback on our ideas for anonymizing social media images.

In the spirit of collegial dialogue about ethics and social media images, this paper's aim has been to share our collaborative process for troubleshooting ethical and methodological issues that emerged in the process of anonymizing online images. Our findings add to ongoing cross-disciplinary conversations about ethical and methodological practices in online research.

About the authors

Nicole Taylor is an associate professor of anthropology at Texas State University. Her research interests include ethics and internet research, social media practices of young adults, teasing and bullying in schools, childhood obesity, and body image concerns among youth.
E-mail: ntaylor [at] txstate [dot] edu

Louie Dean Valencia-García is an associate professor of digital history at Texas State University. His research interests include the history of antiauthoritarian youth culture in Spain, HIV/AIDS in Europe, community organizing, technological innovation and information distribution, and scientific research.
E-mail: lvalencia [at] txstate [dot] edu

Angela K. VandenBroek is an assistant professor of anthropology at Texas State University. Her research interests include business and design anthropology, science and technology studies, entrepreneurship, and innovation culture.
E-mail: akvbroek [at] txstate [dot] edu

Ashley Stinnett is an independent researcher and consultant. Her research interests include visual anthropology, applied anthropology, community based participatory research methods, and linguistic anthropology.
E-mail: astinnet [at] arizona [dot] edu

Alejandro Allen is an independent researcher. He has a M.A. in anthropology from Texas State University. His research interests include food insecurity among college students and addiction.
E-mail: alejandroallenv [at] gmail [dot] com

Acknowledgements

We thank the National Science Foundation for supporting this research (award #1745222) and Texas State University's Institutional Review Board for providing input and feedback on preliminary findings. We are also grateful to *First Monday's* anonymous peer reviewers for their helpful comments.

Notes

1. Horst and Miller, 2012, p. 149.

2. Pearce, *et al.*, 2020, p. 166.

3. https://en.wikipedia.org/wiki/Reverse_image_search.

4. Inconsistency regarding privacy and searchability across social media platforms is due to the many ways in which Google's Web crawlers — "spiderbots" that scour the Web, going link to link, indexing information on the Internet — can be blocked or turned away. Some Web sites use a "rel=nofollow" code, which asks Web crawlers to not follow a link — a sort of a courtesy code; others use a "noindex" code which asks Web crawlers not to index. Instagram's Web app mostly uses JavaScript, which functions differently than HTML code, making it difficult for Google's Web crawlers to recognize links. A platform like Facebook can also require an account to enter the platform. This effectively creates a "paywall" which blocks access to anyone without an account. On Facebook, users can make their profiles public, but can also elect to block Web crawlers. In this way, a person can search for someone from within the platform and locate them, but still not be able to find them on Google. Users can also choose to make their profiles public and searchable by Web crawlers. Because of a variety of circumstances, from privacy settings, to coding, to "paywalls," it is often unpredictable which images and URLs will be indexed in Google. On Twitter, even if users elect to make their "tweets" private, the platform still makes public names, profile photos and banners (large custom images at the top of user pages). These elements are thus indexable by Google.

5. https://www.photoshopessentials.com/photo-effects/photo-to-color-pencil-sketch-with-photoshop-cc/

6. boyd, 2016, pp. 7–8.

7. franzke, *et al.*, 2020, p. 23.

References

C. Abidin, 2018. *Internet celebrity: Understanding fame online*. Bingley: Emerald.
doi: https://doi.org/10.1108/9781787560765, accessed 22 March 2023.

R.S. Arora and A. Elgammal, 2012. "Towards automated classification of fine-art painting style: A comparative study," *Proceedings of the 21st International Conference on Pattern Recognition (ICPR2012)*, pp. 3,541–3,544.
doi: https://doi.org/10.7282/T3XP73QP, accessed 22 March 2023.

A. Beaulieu and A. Estalella, 2012. "Rethinking research ethics for mediated settings," *Information, Communication & Society*, volume 15, number 1, pp. 23–42.
doi: https://doi.org/10.1080/1369118X.2010.535838, accessed 22 March 2023.

K. Beninger, A. Fry, N. Jago, H. Lepps, L. Nass, and H. Silvester, 2014. "Research using social media users' views," *National Centre for Research Methods*, at https://www.bl.uk/collection-items/research-using-social-media-users-views, accessed 22 March 2023.

S.M. Blalock and S. Farrar, 2019. "Whitman and Dickinson," *American Literary Scholarship*, volume 2017, number 1, pp. 49–66.
doi: https://doi.org/10.1215/00659142-7328785, accessed 22 March 2023.

T. Boellstorff, B. Nardi, C. Pearce, and T.L. Taylor, 2012. *Ethnography and virtual worlds: A handbook of methods*. Princeton, N.J.: Princeton University Press.

P. Bourgois and J. Schonberg, 2009. *Righteous dopefiend*. Berkeley: University of California Press.

d. boyd, 2016. "Untangling research and practice: What Facebook's 'emotional contagion' study teaches us," *Research Ethics*, volume 12, number 1, pp. 4–13.
doi: https://doi.org/10.1177/1747016115583379, accessed 22 March 2023.

d. boyd and K. Crawford, 2012. "Critical questions for big data: Provocations for a cultural, technological, and scholarly phenomenon," *Information, Communication & Society*, volume 15, number 5, pp. 662–697.
doi: https://doi.org/10.1080/1369118X.2012.678878, accessed 22 March 2023.

M.A. Carney, 2015. *The unending hunger: Tracing women and food insecurity across borders*. Berkeley: University of California Press.

A. Clark, 2012. "Visual ethics in a contemporary landscape," In: S. Pink (editor). *Advances in visual methodology*. Los Angeles, Calif..: Sage, pp. 17–36.
doi: https://doi.org/10.4135/9781446250921, accessed 22 March 2023.

C. Cömert, M. Özbayoğlu, and C. Kasnakoğlu, 2021. "Painter prediction from artworks with transfer learning," *2021 Seventh International Conference on Mechatronics and Robotics Engineering (ICMRE)*, pp. 204–208.
doi: https://doi.org/10.1109/ICMRE51691.2021.9384828, accessed 22 March 2023.

S. Cox, S. Drew, M. Guillemin, C. Howell, D. Warr, and J. Waycott, 2014. "Guidelines for ethical visual research methods," *University of Melbourne*, at
https://edisciplinas.usp.br/pluginfile.php/4434501/mod_resource/content/1/Ethical%20guidelines%20for%20visual%20ethnography.pdf, accessed 22 March 2023.

J.L. Davis and N. Jurgenson, 2014. "Context collapse: Theorizing context collusions and collisions," *Information, Communication & Society*, volume 17, number 4, pp. 476–485.
doi: https://doi.org/10.1080/1369118X.2014.888458, accessed 22 March 2023.

J. De Leon, 2015. *The land of open graves: Living and dying on the migrant trail*. Berkeley: University of California Press.

A. Elgammal, B. Liu, D. Kim, M. Elhoseiny, and M. Mazzone, 2018. "The shape of art history in the eyes of the machine," *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, number 1, pp. 2,183–2,191.
doi: https://doi.org/10.1609/aaai.v32i1.11894, accessed 22 March 2023.

N. Emmel and A. Clark, 2011. "Learning to use visual methodologies in our research: A dialogue between two researchers," *Forum: Qualitative Social Research*, volume 12, number 1.
doi: https://doi.org/10.17169/fqs-12.1.1508, accessed 22 March 2023.

a.s. franzke, A. Bechmann, M. Zimmer, C. Ess, and the Association of Internet Researchers (AOIR), 2020. "Internet research: Ethical guidelines 3.0," at https://aoir.org/reports/ethics3.pdf, accessed 22 March 2023.

I. Gershon, 2010. *The breakup 2.0: Disconnecting over new media*. Ithaca, N.Y.: Cornell University Press.

M.H. Goodwin, 2006. *The hidden life of girls: Games of stance, status, and exclusion*. Malden, Mass.: Blackwell.
doi: https://doi.org/10.1002/9780470773567, accessed 22 March 2023.

K. Hall, 2016. “Selfies and self-writing: Cue card confessions as social media technologies of the self,” *Television & New Media*, volume 17, number 3, pp. 228–242.
doi: https://doi.org/10.1177/1527476415591221, accessed 22 March 2023.

E. Hargittai and A. Marwick, 2016. “What can I really do? Explaining the privacy paradox with online apathy,” *International Journal of Communication*, volume 10, pp. 3,737-3,757, and at https://ijoc.org/index.php/ijoc/article/view/4655, accessed 22 March 2023.

K.G. Heider, 2006. *Ethnographic film*. Revised edition. Austin: University of Texas Press.

H.A. Horst and D. Miller (editors), 2012. *Digital anthropology*. London: Berg.

S. Khamis, L. Ang, and R. Welling, 2017. “Self-branding, ‘micro-celebrity’ and the rise of social media influencers,” *Celebrity Studies*, volume 8, number 2, pp. 191–208.
doi: https://doi.org/10.1080/19392397.2016.1218292, accessed 22 March 2023.

S. Lomborg, 2013. “Personal Internet archives and ethics,” *Research Ethics*, volume 9, number 1, pp. 20–31.
doi: https://doi.org/10.1177/1747016112459450, accessed 22 March 2023.

S. Lomborg and A. Bechmann, 2014. “Using APIs for data collection on social media,” *Information Society*, volume 30, number 4, pp. 256–265.
doi: https://doi.org/10.1080/01972243.2014.915276, accessed 22 March 2023.

L. Manderson, M. Davis, C. Colwell, and T. Ahlin, 2015. “On secrecy, disclosure, the public, and the private in anthropology,” *Current Anthropology*, volume 56, number S12, pp. 183–190.
doi: https://doi.org/10.1086/683302, accessed 22 March 2023.

A. Markham, 2012. “Fabrication as ethical practice: Qualitative inquiry in ambiguous Internet contexts,” *Information, Communication & Society*, volume 15, number 3, pp. 334–353.
doi: https://doi.org/10.1080/1369118X.2011.641993, accessed 22 March 2023.

A.N. Markham and E.A. Buchanan, 2015. “Internet research: Ethical concerns,” In: J.D. Wright (editor-in-chief). *International encyclopedia of the social & behavioral sciences*. Second edition. Amsterdam: Elsevier, pp. 606–613.
doi: https://doi.org/10.1016/B978-0-08-097086-8.11027-X, accessed 22 March 2023.

A. Markham and E. Buchanan, 2012. “Ethical decision-making and Internet research: Recommendations from the AoIR Ethics Working Committee,” version 2.0, at http://aoir.org/reports/ethics2.pdf, accessed 22 March 2023.

A.E. Marwick and d. boyd, 2014. “Networked privacy: How teenagers negotiate context in social media,” *New Media & Society*, volume 16, number 7, pp. 1,051–1,067.
doi: https://doi.org/10.1177/1461444814543995, accessed 22 March 2023.

N. Mendoza-Denton, 2008. *Homegirls: Language and cultural practice among Latina youth gangs*. Malden, Mass.: Blackwell.
doi: https://doi.org/10.1002/9780470693728, accessed 22 March 2023.

D. Miller and J. Sinanan, 2017. *Visualising Facebook: A comparative perspective*. London: UCL Press.
doi: https://doi.org/10.2307/j.ctt1mtz51h, accessed 22 March 2023.

H. Nissenbaum, 2011. “A contextual approach to privacy online,” *Daedalus*, volume 140, number 4, pp. 32–48.
doi: https://doi.org/10.1162/DAED_a_00113, accessed 22 March 2023.

A. Patterson, 2018. “YouTube generated video clips as qualitative research data: One researcher’s reflections on the process,” *Qualitative Inquiry*, volume 24, number 10, pp. 759–767.
doi: https://doi.org/10.1177/1077800418788107, accessed 22 March 2023.

W. Pearce, S.M. Özkula, A.K. Greene, L. Teeling, J.S. Bansard, J.J. Omena, and E.T. Rabello, 2020. “Visual cross-platform analysis: Digital methods to research social media images,” *Information, Communication & Society*, volume 23, number 2, pp. 161–180.
doi: https://doi.org/10.1080/1369118X.2018.1486871, accessed 22 March 2023.

S. Pink (editor), 2012. *Advances in visual methodology*. Los Angeles, Calif.: Sage.
doi: https://doi.org/10.4135/9781446250921, accessed 22 March 2023.

S. Ravn, A. Barnwell, and B.B. Neves, 2020. “What is ‘publicly available data’? Exploring blurred public-private boundaries and ethical practices through a case study on Instagram,” *Journal of Empirical Research on Human Research Ethics*, volume 15, numbers 1–2, pp. 40–45.
doi: https://doi.org/10.1177/1556264619850736, accessed 22 March 2023.

K. Raynes-Goldie, 2010. “Aliases, creeping, and wall cleaning: Understanding privacy in the age of Facebook,” *First Monday*, volume 15, number 1, at https://firstmonday.org/article/view/2775/2432, accessed 22 March 2023.
doi: https://doi.org/10.5210/fm.v15i1.2775, accessed 22 March 2023.

K. Strand, S. Marullo, N. Cutforth, R. Stoecker, and P. Donohue, 2003. “Principles of best practice for community-based research,” *Michigan Journal of Community Service Learning*, volume 9, number 3, pp. 5–15, and at http://hdl.handle.net/2027/spo.3239521.0009.301, accessed 22 March 2023.

N. Taylor and M. Nichter, 2022. *A filtered life: Social media use among college students*. New York: Routledge.
doi: https://doi.org/10.4324/9781003182047, accessed 22 March 2023.

K. Tiidenberg, 2018. “Ethics in digital research,” In: U. Flick (editor). *Sage handbook of qualitative data collection*. Thousand Oaks, Calif.: Sage, pp. 466–479.
doi: https://doi.org/10.4135/9781526416070, accessed 22 March 2023.

L. Townsend and C. Wallace, 2016. “Social media research: A guide to ethics,” *University of Aberdeen*, at https://www.gla.ac.uk/media/media_487729_en.pdf, accessed 22 March 2023.

R. Tromble and D. Stockmann, 2017. “Lost umbrellas: Bias and the right to be forgotten in social media research,” In: M. Zimmer and K. Kinder-Kurlanda (editors). *Internet research ethics for the social age: New challenges, cases, and contexts*. New York: Peter Lang, pp. 75–91.
doi: https://doi.org/10.3726/b11077, accessed 22 March 2023.

K. Warfield, J. Hoholuk, B. Vincent, and A.D. Camargo, 2019. “Pics, dicks, tits, and tats: Negotiating ethics working with images of bodies in social media research,” *New Media & Society*, volume 21, number 9, pp. 2,068–2,086.
doi: https://doi.org/10.1177/1461444819837715, accessed 22 March 2023.

R. Wiles, A. Coffey, J. Robinson, and S. Heath, 2012. “Anonymisation and visual images: Issues of respect, ‘voice’ and protection,” *International Journal of Social Research Methodology*, volume 15, number 1, pp. 41–53.
doi: https://doi.org/10.1080/13645579.2011.564423, accessed 22 March 2023.

M. Zimmer and K. Kinder-Kurlanda (editors), 2017. *Internet research ethics for the social age: New challenges, cases, and contexts*. New York: Peter Lang.

Editorial history

Received 19 June 2022; revised 4 November 2022; revised 7 March 2023; accepted 7 March 2023.

This paper is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License.

Ethics and images in in social media research
by Nicole Taylor, Louie Dean Valencia-García, Angela VandenBroek, Ashley Stinnett, and Alejandro Allen.
First Monday, volume 28, number 4 (April 2023).
doi: https://dx.doi.org/10.5210/fm.v28i4.12680